THURRO INDUSTRY PROJECT 05 · AIPI 590

Do faster home sales call cement demand early?

Residential launch-to-sales velocity as a 1–2 month lead on India’s national cement print

Sept 30 progress update · [Team member names]

Our understanding of the problem

When new homes sell faster in a city, builders soon need more cement. We test whether that city-level signal can call the direction of national cement demand 1–2 months before the official number lands.

Practically, this is a **nowcast**: the housing data is available before the cement print for the same months, so the lead comes from release timing as much as from economics.

Velocity metric: absorption = homes sold ÷ (opening unsold stock + new launches), per city per quarter.

City signal, national target

Seven metro cities must be aggregated into one index: equal-weighted vs construction-share-weighted is a design choice we have to test.

Frequency mismatch

Launches and sales are quarterly; cement is monthly. Weekly RERA project feeds are the bridge.

Confounders

Lending rates move housing and industry together (MCLR control); infrastructure capex drives cement on its own.

Target datasets we identified

Dataset	Role	Frequency	Coverage on Thurro	First read
Launches & sales (ANAROCK, 7 cities)	Core signal	Quarterly	Q1 2018 – Q2 2026	Clean; Q4 2025 missing
RERA registrations (Karnataka, Telangana, Haryana)	Monthly bridge	Weekly snapshots	Late 2021 – Sep 2026	Only TG and HR record bookings; scrape gaps
Core Industries Index: cement	Target	Monthly	Old base Apr 2011 – May 2026; new base Apr 2023 – Aug 2026	Two bases splice cleanly → 8 years of target
1-year MCLR (bank-level, group medians)	Control	Monthly	Apr 2016 – Aug 2026	Current, complete

Key find: the glossary lists cement for 2023–2026 only, but an **older base series runs back to 2011**, so the target overlaps the full ANAROCK history instead of ~12 quarters.

EDA 1 — What we can trust in the data

±200

units: the inventory identity holds

Unsold stock = last quarter's stock + launches − sales in every city-quarter (gaps are rounding). That lets us recover net supply for the missing Q4 2025: +14,800 units across 7 cities.

0.996

YoY correlation, old vs new cement base

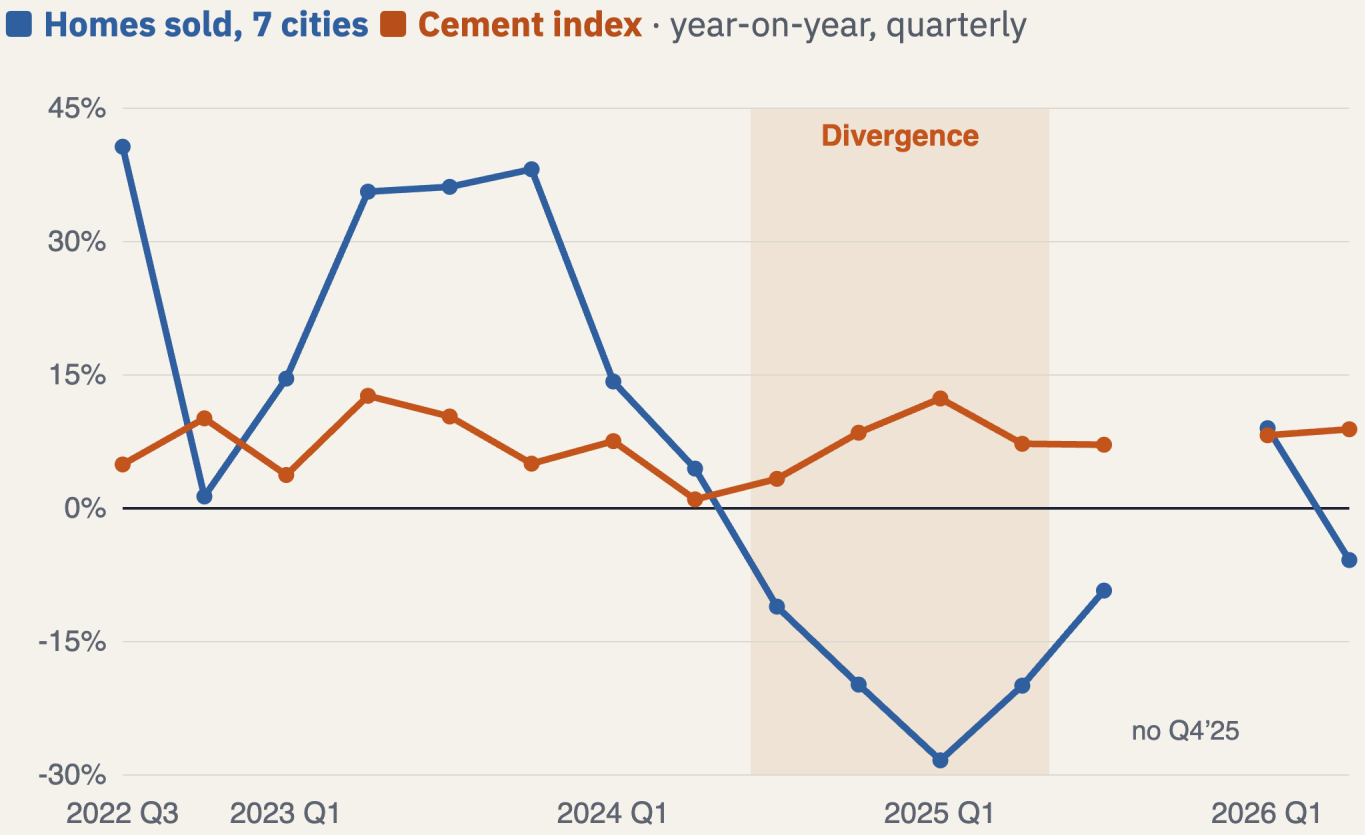
Over 26 overlapping months the two bases differ by a near-constant level ratio (1.705). We splice them into one monthly target, Jan 2018 – Aug 2026.

33/34

quarters present

Q4 2025 is missing for all cities. Q2 2020 launches fell to ~1.4k units from 41k the quarter before, so COVID quarters get masked or dummied.

EDA 2 — Housing and cement co-move, but not yet ahead



0.56

Same-quarter correlation of sales YoY with cement YoY (ex-COVID, 21 quarters)

0.09

Sales one quarter ahead: the quarterly lag alone does not lead

Q3 2024 – Q2 2025: sales fell 11–28% YoY while cement grew 3–12%: non-housing demand dominated.

What success looks like, and how we will evaluate it

First, define “direction” fairly

Month-to-month, the direction of cement YoY growth **flips 74% of the time** (80 months, ex-COVID). A trivial “it will reverse” rule would look brilliant.

On a **3-month YoY** target, it persists 46% and flips 54%: close to a coin toss, so beating naive rules means something.

Primary metric: hit rate on the direction of 3-month cement YoY, called 1 and 2 months before the official release.

Backtest: walk-forward, expanding window, respecting each source’s publication lag.

Baselines to beat: persistence, seasonal naive, cement-only autoregression.

Stretch: the edge survives adding MCLR; compare equal vs construction-share weights.

Success: beat the best baseline out of sample, with a significance check given ~50 months.

Roadmap and next steps

PHASE 1

Monthly panel

Clean Telangana and Haryana RERA (completeness filter, de-dup), monthly booking deltas, spliced cement target, publication-lag calendar.

PHASE 2

Nowcast models

Bridge quarterly ANAROCK to monthly with RERA; direction models; walk-forward backtest vs baselines.

PHASE 3

Aggregation & controls

Equal vs construction-share city weights; MCLR control; ex-COVID robustness.

PHASE 4

Data product

Monthly “cement direction call” with a confidence score and the cities driving it; final write-up.

Now: starting Phase 1 · [add target dates per phase]

Questions for the Thurro team

1. **Release timing:** can we get ANAROCK publish dates and Core Index vintages? History appears bulk-loaded, so we cannot rebuild what was known when.
2. **Q4 2025 ANAROCK:** is it coming, or known to be missing?
3. **Telangana RERA gaps** (May 2022 – Sep 2023, mid-2026): known issue? Is there a completeness flag?
4. **Construction-share weights:** does Thurro hold city or state construction or cement-consumption data?
5. **Benchmark:** is “1–2 months early” measured vs the release date or the reference month, and is the Core Index cement series the intended target?